Tutorial 13

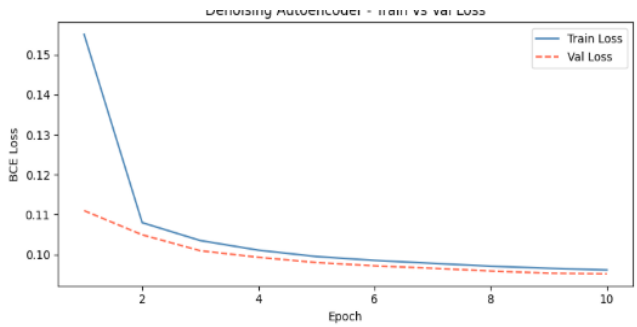
Task 1 – Basic Denoising Autoencoder

Architecture

- Encoder: Conv2d(1→32) → ReLU → MaxPool, Conv2d(32→64) → ReLU → MaxPool → 64×7×7
- Decoder: Conv2d(64→64) → Upsample → Conv2d(64→32) → Upsample → Conv2d(32→1) → Sigmoid
- Loss: **BCE**, Optimizer: **Adam (lr=0.001)**, Epochs: 10, Batch: 128, Device: **GPU**

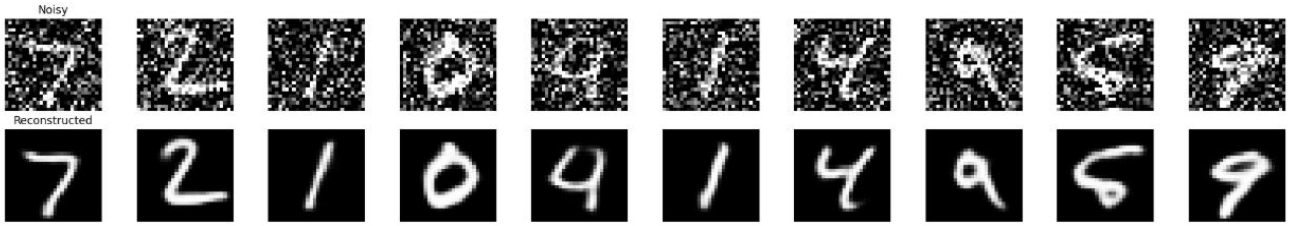
Results

```
Epoch [1/10]  Train Loss: 0.1551  Val Loss: 0.1110
Epoch [2/10]  Train Loss: 0.1080  Val Loss: 0.1050
Epoch [3/10]  Train Loss: 0.1035  Val Loss: 0.1010
Epoch [4/10]  Train Loss: 0.1011  Val Loss: 0.0993
Epoch [5/10]  Train Loss: 0.0995  Val Loss: 0.0980
Epoch [6/10]  Train Loss: 0.0986  Val Loss: 0.0972
Epoch [7/10]  Train Loss: 0.0978  Val Loss: 0.0966
Epoch [8/10]  Train Loss: 0.0971  Val Loss: 0.0959
Epoch [9/10]  Train Loss: 0.0966  Val Loss: 0.0954
Epoch [10/10]  Train Loss: 0.0962  Val Loss: 0.0952
```



Analysis

- Train loss decreased from **0.1551 → 0.0962** and val loss from **0.1110 → 0.0952** across 10 epochs, showing steady convergence.
- Val loss stays consistently below train loss throughout all epochs, which confirms the model is generalizing well and there is no overfitting.
- The loss drops steeply in the first 2–3 epochs, then flattens out which typical behavior as the model quickly learns the dominant noise patterns before fine-tuning.
- All 10 reconstructed digits (7, 2, 1, 0, 4, 1, 4, 9, 5, 9) are clean and sharp despite heavy Gaussian noise (factor=0.5), demonstrating the encoder has learned a noise-invariant compressed representation.
- The simple 2-layer encoder is sufficient for this task, MNIST digits have low complexity, so a shallow network can already capture the essential structure needed for denoising.



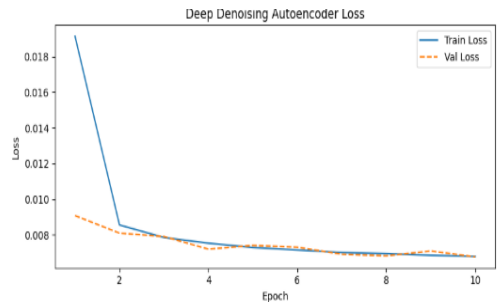
Task 2 – Deep Autoencoder

Architecture

- 3-stage encoder with double Conv layers + **BatchNorm** at each stage (32→64→128 filters).
- Symmetric decoder with Upsample layers restoring 1×28×28 output.
- Loss: **BCE**, Optimizer: **Adam (lr=0.001)**, Epochs: 10, Batch: 128

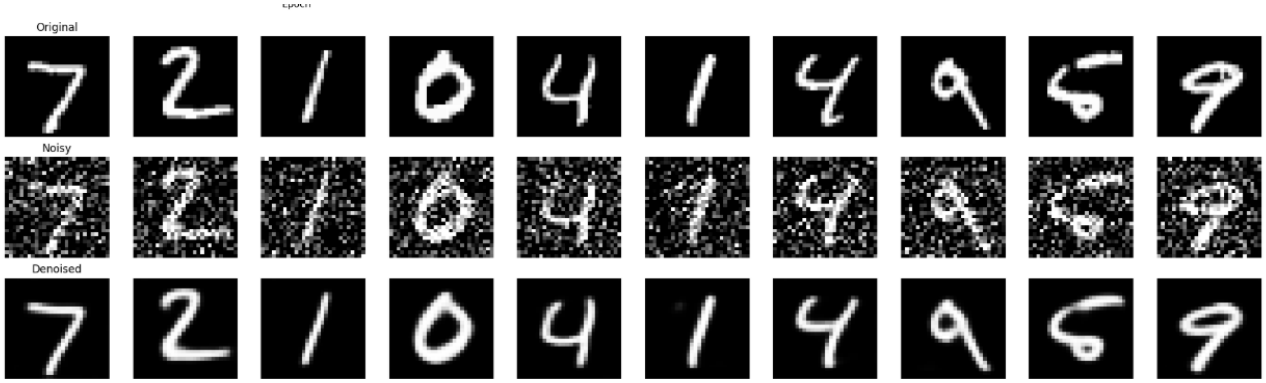
Results

```
Epoch [1/10]  Train Loss: 0.0191  Val Loss: 0.0091
Epoch [2/10]  Train Loss: 0.0086  Val Loss: 0.0081
Epoch [3/10]  Train Loss: 0.0079  Val Loss: 0.0079
Epoch [4/10]  Train Loss: 0.0075  Val Loss: 0.0072
Epoch [5/10]  Train Loss: 0.0073  Val Loss: 0.0074
Epoch [6/10]  Train Loss: 0.0072  Val Loss: 0.0073
Epoch [7/10]  Train Loss: 0.0070  Val Loss: 0.0069
Epoch [8/10]  Train Loss: 0.0070  Val Loss: 0.0068
Epoch [9/10]  Train Loss: 0.0069  Val Loss: 0.0071
Epoch [10/10]  Train Loss: 0.0068  Val Loss: 0.0068
```



Analysis

- Train loss dropped from **0.0191 → 0.0068** and val loss from **0.0091 → 0.0068** , roughly **14× lower** than the basic DAE, showing a major improvement from added depth and BatchNorm.
- The large drop in the first epoch (0.0191 → 0.0086) is due to the deeper network rapidly learning multi-scale features that the shallow DAE could not capture.
- Train and val losses converge to nearly the same value (both 0.0068 by Epoch 10), confirming the model generalizes well with no overfitting despite having more parameters.
- BatchNorm plays a key role here , it normalizes activations at each layer, preventing internal covariate shift, stabilizing gradients, and allowing the deeper network to train reliably without vanishing gradients.
- Reconstructed digits are noticeably sharper than Task 1 , fine stroke details, digit curves, and edges are better preserved, which directly reflects the richer feature hierarchy learned by the deeper encoder.



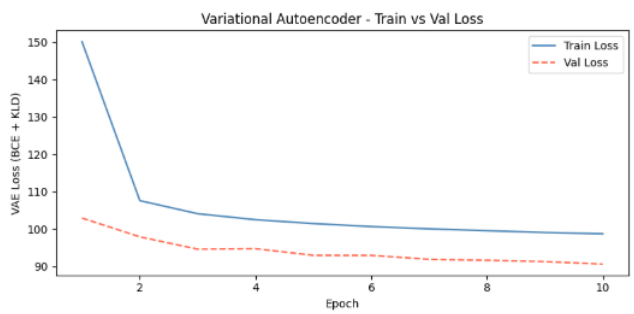
Task 3 : Variational Autoencoder (VAE)

Architecture

- Encoder outputs **mean (μ) and log-variance (logσ²)** with a latent dimension of 16 instead of a fixed vector.
- Uses **reparameterization trick**: $z = \mu + \epsilon \cdot \sigma z$, enabling differentiable sampling.
- Decoder maps latent vector (16 → 6272) and reshapes it to reconstruct **28×28 images using convolutional upsampling**.
- Trained with **BCE loss + KL divergence** to balance reconstruction quality and latent space regularization.
- Model has **492,577 parameters** and was trained on **CPU**.

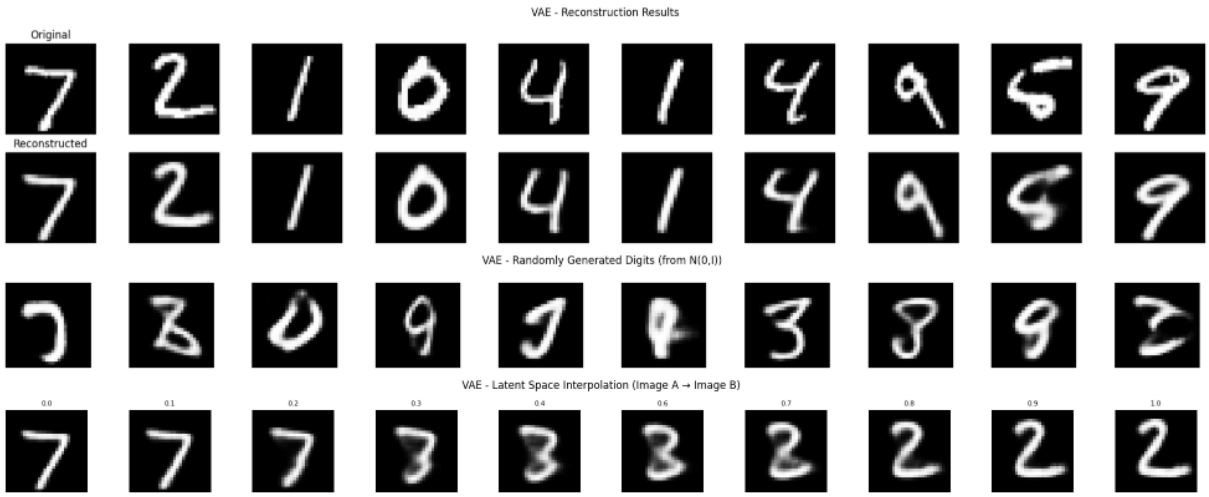
Results

```
Latent Dimension: 16
Total Trainable Parameters: 492,577
Epoch [1/10]  Train Loss: 150.1598  Val Loss: 102.9161
Epoch [2/10]  Train Loss: 107.5850  Val Loss: 97.8933
Epoch [3/10]  Train Loss: 104.0847  Val Loss: 94.5828
Epoch [4/10]  Train Loss: 102.4804  Val Loss: 94.7103
Epoch [5/10]  Train Loss: 101.4464  Val Loss: 92.9371
Epoch [6/10]  Train Loss: 100.6385  Val Loss: 92.9165
Epoch [7/10]  Train Loss: 100.0260  Val Loss: 91.8756
Epoch [8/10]  Train Loss: 99.5404  Val Loss: 91.6021
Epoch [9/10]  Train Loss: 99.0461  Val Loss: 91.2559
Epoch [10/10]  Train Loss: 98.7067  Val Loss: 90.5920
```



Analysis

- Train loss decreased from 150.16 → 98.71 and validation loss from 102.92 → 90.59 over 10 epochs. The higher values compared to earlier models are expected because the loss includes both BCE and KL divergence, making it not directly comparable.
- Validation loss remains consistently lower than training loss, indicating good generalization and no overfitting.
- Reconstructions are slightly blurrier but smoother, which is a known property of VAEs due to probabilistic sampling in the latent space.
- The model can generate new digit images by sampling which is not possible with standard autoencoders.
- The latent space supports smooth interpolation between digits, producing gradual transitions (e.g., 7 → 2), due to KL regularization enforcing a structured and continuous representation.



Comparison

Model	Train Loss	Val Loss	Key Advantage
Basic DAE	0.0962	0.0952	Simple, fast
Deep AE	0.0068	0.0068	Best reconstruction
VAE	98.71*	90.59*	Generative

Conclusion

- The Basic DAE (Task 1) removes Gaussian noise using a simple CNN encoder–decoder, reaching val BCE ≈ 0.0952. It is lightweight and effective for basic denoising on MNIST.
- The Deep Autoencoder (Task 2) improves reconstruction using deeper Conv + BatchNorm layers, achieving much lower loss (≈ 0.0068) and sharper outputs with stable training.
- The VAE (Task 3) learns a probabilistic latent space using KL divergence. It produces slightly blurrier reconstructions but enables image generation and smooth interpolation.
- Overall: Basic DAE = simplicity, Deep AE = best reconstruction, VAE = generative capability + structured latent space.